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EMC FILTER FOR ELECTROMAGNETIC REGULATION OF CONVERTER AND MANUFACTURING METHOD THEREOF

Abstract

Provided is an electro-magnetic compatibility (EMC) filter including a lower bobbin having a U-shaped cross-sectional shape, a lower core including a magnetic material having a U-shaped cross-sectional shape and disposed on the lower bobbin, a bus bar disposed on the lower core, an upper bobbin having a hollow inside, having a bexahedral shape with one side open, and configured to cover an upper portion of the lower bobbin, and an upper core including a magnetic material having a plate-like shape, disposed in an internal space of the upper bobbin, and disposed on the lower core (U core) to cover the bus bar with a gap maintained by the bus bar between the upper and lower cores when the lower bobbin and the upper bobbin are coupled to each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is the continuation application of U.S. patent application Ser. No. 17/530,756 filed on Nov. 19, 2021, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2020-0157098, filed on Nov. 20, 2020, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to an external EMC filter for satisfying electromagnetic wave regulation of a converter in a vehicle.

BACKGROUND

[0003] A power conversion device for converting and controlling electric energy into various types of power required by each electric device is installed in a vehicle. A typical example of such a power conversion device is a converter (e.g., a DC-DC converter).

[0004] An electro-magnetic compatibility (EMC) filter is connected to an output terminal of a converter to reduce electromagnetic noise occurring in an output of the converter.

[0005] An EMC filter of a related art includes a bus bar, a core, and a bobbin. According to the EMC filter of the related art, the bobbin accommodates the core, and the core accommodated in the bobbin surrounds the bus bar through which a large current flows. In the case of the core, an air gap is formed inside the core to prevent saturation by a large current.

[0006] In the case of the EMC filter of the related art, the bobbin is manufactured to have a special shape to maintain a gap inside the core. However, since the bobbin is formed of a plastic material, it is vulnerable to external vibration and shock.

[0007] When the bobbin is damaged by external vibration and impact, it may be difficult to maintain a gap inside the core, and thus, there is a problem in that saturation of the large current flowing in the bus bar cannot be prevented.

[0008] In addition, according to a fringing effect, heat is generated in the bus bar by a magnetic field (fringing field) generated in the gap inside the core, thereby increasing a temperature of the bus bar.

SUMMARY

[0009] Accordingly, the present disclosure provides an electro-magnetic compatibility (EMC) filter capable of minimizing a temperature rise of a bus bar due to a fringing field generated in a gap of a core and being robust to external vibration and impact, and a manufacturing method thereof.

[0010] The above and other objects, advantages and features of the present disclosure, and a method of achieving them will become apparent with reference to the embodiments described below in detail in conjunction with the accompanying drawings.

[0011] In one general aspect, an electro-magnetic compatibility (EMC) filter includes: a lower bobbin having a U-shaped cross-sectional shape; a lower core including a magnetic material having a U-shaped cross-sectional shape and disposed on the lower bobbin; a bus bar disposed on the lower core; an upper bobbin having a hollow inside, having a hexahedral shape with one side open, and configured to cover an upper portion of the lower bobbin; and an upper core including a magnetic material having a plate-like shape, disposed in an internal space of the upper bobbin, and disposed on the lower core (U core) to cover the bus bar with a gap maintained by the bus bar between the upper and lower cores when the lower bobbin and the upper bobbin are coupled to each other.

[0012] The bus bar may be configured to extend to bypass the gap so as not to overlap the gap.

[0013] The bus bar may include a first bus bar configured to extend below a height level of the gap so as not to overlap the gap; and second and third bus bars configured to extend from respective upper end surfaces of two ends of the first bus bar in opposite directions.

[0014] A height of the lower core may be a thickness of the second bus bar or the third bus bar, or the height of the lower core may be designed by a following equation: (the height of the lower core = $2 \times$ a thickness of the first bus bar - the gap).

[0015] The EMC filter may further include: a heat dissipation material applied to a portion of the bus bar and a portion of the lower core not covered by the bus bar and exposed upwardly. Here, a portion of the bus bar may be a surface of the first bus bar.

[0016] In another general aspect, a method of manufacturing an electro-magnetic compatibility (EMC) filter includes: attaching a lower core having a U-shaped cross-sectional shape to a lower bobbin having a U-shaped cross-sectional shape; attaching a bus bar to the lower core; applying a heat dissipation material to a portion of the bus bar and the lower core exposed upwardly without being covered by the bus bar; attaching an upper core to a lower surface forming an internal space of the upper bobbin; and coupling the lower bobbin and the upper bobbin such that the lower core and the upper core encase the bus bar with a gap between the lower core and the upper core maintained by the bus bar.

[0017] In another general aspect, an electro-magnetic compatibility (EMC) filter includes: a lower bobbin having a U-shaped cross-sectional shape; a lower core (U core) having a magnetic material, having a U-shaped cross-sectional

shape, and disposed on the lower bobbin; a bus bar disposed on the lower core; an upper bobbin having a plate-like shape and configured to cover an upper portion of the lower bobbin; and an upper core (I core) having a magnetic material, having a plate-like shape, disposed on a lower surface of the upper bobbin, and disposed on the lower core (U core) with a gap maintained by the bus bar when the lower bobbin and the upper bobbin are coupled to each other. [0018] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is a view showing an EMC filter mounted on a converter according to an embodiment of the present disclosure.

[0020] FIG. 2 is a perspective view of an EMC filter according to an embodiment of the present disclosure.

[0021] FIG. 3A and 3B are front views and a top view of the EMC filter shown in FIG. 2 together.

[0022] FIG. 4 is an exploded perspective view of the EMC filter shown in FIG. 2.

[0023] FIG. 5 is a cross-sectional view of the EMC filter, taken along line I-I' shown in FIG. 2.

[0024] FIG. 6 is a cross-sectional view of the EMC filter, taken along line II-II' shown in FIG. 2.

[0025] FIGS. 7 to 12 are views showing a manufacturing process of an EMC filter according to an embodiment of the present disclosure.

[0026] FIG. 13 is a perspective view of an EMC filter according to another embodiment of the present disclosure.

[0027] FIG. 14 is a cross-sectional view of the EMC filter, taken along line I-I' shown in FIG. 13.

[0028] FIG. 15 is a cross-sectional view of the EMC filter, taken along line II-II' shown in FIG. 13.

DETAILED DESCRIPTION OF EMBODIMENTS

[0029] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

[0030] The advantages, features and aspects of the present disclosure will become apparent from the following description of the embodiments with reference to the accompanying drawings, which is set forth hereinafter. The present disclosure may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art. In this disclosure, when an element is described as being connected to another element, the element may be directly connected to the other element, or a third element may be interposed therebetween. Also, in the drawings, a shape or a size of each element is exaggerated for convenience of a description and clarity, and elements irrelevant to a description are omitted. Like reference numerals refer to like elements throughout. The terms of a singular form may include plural forms unless referred to the contrary. The meaning of 'comprise', 'include', or 'have' specifies a property, a region, a fixed number, a step, a process, an element and/or a component but does not exclude other properties, regions, fixed numbers, steps, processes, elements and/or components.

[0031] FIG. 1 is a view showing an EMC filter mounted on a converter according to an embodiment of the present disclosure.

[0032] Referring to FIG. 1, an EMC filter **100** is formed on a cooling passage **210** in a housing **200** (hereinafter, referred to as an 'outer housing') forming an outer periphery of a converter (e.g., a DC-DC converter).

[0033] As shown in FIG. 1, since the EMC filter **100** is directly installed on the cooling passage **210** in the outer housing **200** of the converter, the bus bar in the EMC filter **100** may be efficiently cooled as described hereinafter.

[0034] FIG. 2 is a perspective view of an EMC filter according to an embodiment of the present disclosure, FIG. 3A and 3B are front views and a top view of the EMC filter shown in FIG. 2 together, FIG. 4 is an exploded perspective view of the EMC filter shown in FIG. 2, FIG. 5 is a cross-sectional view of the EMC filter, taken along line I-I' shown in FIG. 2, and FIG. 6 is a cross-sectional view of the EMC filter, taken along line II-II' shown in FIG. 2.

[0035] Referring to FIGS. 2 to 6, the EMC filter **100** includes bobbins **110** and **130**, cores **120** and **140** and a bus bar **150**.

[0036] The bobbins **110** and **130** are configured to include a lower bobbin **110** and an upper bobbin **130** covering an upper portion of the lower bobbin **110**.

[0037] As shown in FIG. 4, for example, the lower bobbin **110** may be formed to have a U-shaped cross-sectional structure, and the material may be, for example, a plastic material and may be molded to have a U-shaped cross-sectional structure by an injection molding method.

[0038] The upper bobbin **130** has one side open and is formed in a hexahedral shape with an empty inside, may be formed of the same material as that of the lower bobbin **110**, and may be molded to have a hexahedral shape with one side open and an inside empty by an injection molding method.

[0039] The lower core **120** (or a U core) is disposed on the lower bobbin **110**. Here, the lower core **120** is also formed to have a U-shaped cross-sectional structure so as to be disposed on the lower bobbin **110** having a U-shaped cross-sectional structure.

[0040] The lower core **120** (or a U core) is formed of a magnetic material, and the magnetic material may be, for example, a ferrite-based material.

[0041] In an internal space of the upper bobbin **130**, the upper core **140**, (or an I core in FIGS. 5 and 6) is disposed. In view of the EMC filter **100** of FIGS. 2, 3, and 4, the upper core **140** (I-core) disposed in the internal space of the upper bobbin **130** is not visible, and thus, the upper core **140** (I core) is not illustrated in FIGS. 2, 3, and 4, and illustrated in FIGS. 5 and 6, instead.

[0042] As shown in FIG. 6, the upper core **140** is formed in a plate-like shape, unlike the lower core **120** having a U-shaped cross-sectional structure.

[0043] The upper core **140** (I core) may also be formed of a magnetic material like the lower core **120** (U core).

[0044] When the lower bobbin **110** and the upper bobbin **130** are coupled to each other, a preset gap (G in FIGS. 5 and 6) is formed between the lower core **120** and the upper core **140**. The presence of the gap (G in FIG. 6) is to prevent saturation of a large current flowing in the bus bar **150**, which will be described below.

[0045] The bus bar **150** is disposed on the lower core **120** (U core). Unlike the related art in which a bobbin is manufactured in a special shape to design the gap, in the present disclosure, the gap (in FIGS. 5 and 6) G is maintained by the bus bar **150** formed of a hard metal material.

[0046] Since the gap (G in FIGS. 5 and 6) is maintained by the bus bar **150** formed of a hard metal material, the gap (G in FIG. 6) may be maintained even with strong external vibrations and shocks.

[0047] The bus bar **150** disposed on the lower core **120** (U core) includes first to third bus bars **152**, **154**, and **156** being integrally formed.

[0048] The first bus bar **152** is disposed on the lower core **120** (U core), and extends in a straight line under the gap (G in FIGS. 5 and 6) so as not to overlap the gap (G of FIGS. 5 and 6) formed between the lower core (U core) **120** and the upper core (I core) **140**.

[0049] The second and third bus bars **154** and **156** extend in a straight line in opposite directions from upper end surfaces of both ends of the first bus bar **152**, and when manufacturing of the EMC filter **100** is completed by coupling the lower bobbin **110** and the upper bobbin **130**, the lower bobbin **110** and the upper bobbin **130** are designed to extend to the outside of a coupled assembly.

[0050] The second and third bus bars **154** and **156** are respectively connected to an output terminal (not shown in FIG. 1) formed in the housing (**200** in FIG. 1) of the converter disposed therebelow, whereby the EMC filter **100** filters electromagnetic noise occurring at an output terminal of the converter.

[0051] The bus bar **150** including the first to third bus bars **152**, **154**, and **156** may extend to bypasses the gap (G in FIGS. 5 and 6) so as not to overlap the gap (G in FIGS. 5 and 6) and may be designed to be less affected by a magnetic field (fringing field) occurring in the gap (G in FIGS. 5 and 6), thereby minimizing an increase in temperature of the bus bar **150** caused by the magnetic field (fringing field).

[0052] Of course, as shown in FIG. 5, the ends A and B of the second and third bus bars formed at the upper end surfaces of both ends of the first bus bar **152** overlap the gap (G of FIGS. 5 and 6), but the degree to which the ends A and B of the second and third bus bars and the gap (G in FIGS. 5 and 6) overlap is not significantly large to be affected by the magnetic field (fringing field).

[0053] The bus bar **150** may be formed of a highly conductive metal material. Metal materials are harder than plastic materials. In the present disclosure, as shown in FIGS. 5 and 6, the gap (G in FIGS. 5 and 6) formed between the upper core **120** (U core) and the lower core **140** (I core) is maintained using the bus bar **150** formed of a rigid material.

[0054] Accordingly, the gap (G in FIGS. 5 and 6) may be constantly maintained even with strong external vibrations and shocks.

[0055] Meanwhile, a height (H in FIGS. 4 and 6) of the lower core **120** (U core) according to an embodiment of the present disclosure is designed according to a thickness (B in FIGS. 4 and 6) of the first bus bar **152**.

[0056] Here, the height H of the lower core **120** (U core) may be a thickness of the second and third bus bars **154** and **156**. In this case, the thicknesses of the second and third bus bars **154** and **156** are equal, and the thickness of the first bus bar **152** (B in FIGS. 4 and 6) may be different from the thickness of the second bus bar **154**. In this embodiment, it is assumed that the thickness of the first bus bar **152** (B in FIGS. 4 and 6) is different from the thickness of the second and third bus bars **154** and **156**.

[0057] In this embodiment, the height H of the lower core **120** (U core) may be designed by the following equation.

[00001]

$$\text{Height of lower core (H in FIGS. 4 and 6)} = 2 \times \text{thickness of first bus bar 152 (B in FIGS. 4 and 6)} - \text{gap (G in FIGS. 5 and 6)} \quad [\text{Equation 1}]$$

[0058] FIGS. 7 to 12 are views showing a manufacturing process of an EMC filter according to an embodiment of the present disclosure.

[0059] First, referring to FIG. 7, the lower bobbin **110** having a U-shaped cross-sectional structure is prepared, and the lower core **120** having a U-shaped cross-sectional structure is seated on the lower bobbin **110** through a bonding process.

[0060] Next, referring to FIG. 8, the bus bars **150** (**152**, **154**, and **156**) are seated on the lower core **120** (U core) through a bonding process.

[0061] Next, referring to FIG. 9, a process of applying a heat dissipation material **60** is applied to the first bus bar **152**

constituting the bus bars **150** (**152**, **154**, and **156**) and the lower core **120** (U core) not covered by the bus bars **150** (**152**, **154**, and **156**) but exposed upwardly. Here, the heat dissipation material **60** may be, for example, thermal grease.

[0062] Next, referring to FIG. **10**, the upper bobbin **130** is prepared, and the upper core **140** is seated on a bottom surface forming the internal space of the upper bobbin **130** through a bonding process. Here, the process of FIG. **10** may be performed simultaneously with the process of FIG. **7**.

[0063] Next, referring to FIGS. **11A**, **11B** and **12**, the lower bobbin **110** and the upper bobbin **130** are coupled, and the upper bobbin **130** covers the first bus bar **152** and the lower core **120** to which the heat dissipation material **60** is applied.

[0064] According to the coupling process of the lower bobbin **110** and the upper bobbin **130**, the lower core **120** and the upper core **140** encase the bus bars **150** with the preset gap (G in FIGS. **5** and **6**) therebetween. At this time, the first bus bars **152** constituting the bus bars **150** (**152**, **154**, and **156**) are disposed below the gap (G in FIGS. **5** and **6**), whereby the bus bars **150** (**152**, **154**, and **156**) extend in a structure bypassing the gap (G in FIGS. **5** and **6**).

[0065] In this manner, as the bus bars **150** (**152**, **154**, and **156**) extend in the structure bypassing the gap (G in FIGS. **5** and **6**) as a whole, and overlap between the bus bars **150** (**152**, **154**, and **156**) and the gap (G in FIGS. **5** and **6**) is minimized, so that the bus bars **150** (**152**, **154**, and **156**) are less affected by the magnetic field (fringing effect) occurring in the gap (G in FIGS. **5** and **6**). Therefore, it is possible to minimize an increase in the temperature of the bus bars **150** (**152**, **154**, **156**) due to the magnetic field (fringing effect).

[0066] In addition, since the bus bars **150** (**152**, **154**, and **156**) of a hard material such as a metal material maintain (or support) the gap (G in FIGS. **5** and **6**), the gap (G in FIGS. **5** and **6**) may be constantly maintained even with external vibrations and shocks.

[0067] FIG. **13** is a perspective view of an EMC filter according to another embodiment of the present disclosure, FIG. **14** is a cross-sectional view of the EMC filter, taken along line I-I' shown in FIG. **13**, and FIG. **15** is a cross-sectional view of the EMC filter, taken along line II-II' shown in FIG. **13**.

[0068] Referring to FIGS. **13** to **15**, an EMC filter **100'** according to another embodiment of the present disclosure includes a lower bobbin **110'**, an upper bobbin **130'**, and a bus bar **150'**.

[0069] The lower bobbin **110'** according to another embodiment may be implemented to have the same structure and function as the lower bobbin **110** described above with reference to FIGS. **2** to **12**, and the bus bar **150'** according to another embodiment is also implemented to have the same structure and function as the bus bar **150** described above with reference to FIGS. **2** to **12**.

[0070] Therefore, the description of the lower bobbin **110'** and the bus bar **150'** according to another embodiment of the present disclosure is replaced with the description of the lower bobbin **110** and the bus bar **150** described above with reference to FIGS. **2** to **12**.

[0071] However, the upper bobbin **130'** according to another embodiment is different from the upper bobbin **130** formed of a hexahedral shape with one side open and an empty inside as described above in that the upper bobbin **130'** has a plate-like shape. In this case, the upper core may be disposed on a lower surface of the upper bobbin **130'**, rather than on a bottom surface forming an internal space of the upper bobbin **130'**, unlike the embodiment described above.

[0072] Except for the shape difference, the upper bobbin **130'** and the upper bobbin **130** described above are implemented to have the same function. Therefore, the description of the upper bobbin **130'** is also replaced with the description of the upper bobbin **130** described above.

[0073] According to the EMC filter of the present invention, a heat dissipation material is applied on a bus bar extending to bypass a gap (gap between an upper core and a lower core) inside the core, thereby minimizing a temperature rise of a bus bar that occurs due to a fringing effect (fringing field) in the gap inside the core.

[0074] In addition, since the EMC filter of the present disclosure is directly installed on a cooling passage in the outer housing of the converter, cooling efficiency of the bus bar is improved.

[0075] In addition, according to the EMC filter of the present disclosure, by maintaining the gap inside the core (the gap between the upper core and the lower core) with a bus bar formed of a hard metal material, the gap inside the core may be maintained even for strong external vibrations and shocks.

[0076] In addition, according to the EMC filter of the present disclosure, as described above, since the bus bar maintains the gap inside the core (the gap between the upper core and the lower core), as in the prior art, the bobbin has a special shape to maintain the gap, the bobbin may be manufactured to have a simple shape, rather than a special shape for maintaining the gap, thereby reducing time and cost required for manufacturing the bobbin.

[0077] A number of exemplary embodiments have been described above. Nevertheless, it will be understood that various modifications may be made. For example, suitable results may be achieved if the described techniques are performed in a different order and/or if components in a described system, architecture, device, or circuit are combined in a different manner and/or replaced or supplemented by other components or their equivalents. Accordingly, other implementations are within the scope of the following claims.

Claims

- 1.** An electro-magnetic compatibility (EMC) filter, comprising: a lower bobbin having a U-shaped cross-sectional shape; a lower core including a magnetic material having a U-shaped cross-sectional shape and disposed on the lower bobbin in a first direction; a bus bar disposed on the lower core, the bus bar including an upper surface and a step portion recessed downwardly from the upper surface; an upper bobbin having a plate-like shape and covering an upper portion of the lower bobbin; and an upper core having a magnetic material, having a plate-like shape, disposed in an internal space of the upper bobbin, and disposed on the lower core, wherein the upper core abuts on the upper surface of the bus bar when the lower bobbin and the upper bobbin are coupled to each other.
 - 2.** The EMC filter of claim 1, wherein a gap between the upper and lower cores is smaller than a depth of the recessed portion of the bus bar in the first direction.
 - 3.** The EMC filter of claim 1, wherein an uppermost surface of the lower core is at a level higher than an upper surface of the recessed portion of the bus bar in the first direction.
 - 4.** The EMC filter of claim 1, wherein the bus bar extends outwardly from both ends of the lower bobbin in a second direction, different from the first direction, away from a gap between the upper and lower cores.
 - 5.** The EMC filter of claim 4, wherein the bus bar includes: a first bus bar extending in the second direction and including the step portion; and second and third bus bars extending in the second direction and disposed on the first bus bar, and wherein the first bus bar partially overlaps the second and third bus bars in the first direction.
 - 6.** The EMC filter of claim 5, wherein the first, second, and third bus bars are integrally formed with each other.
 - 7.** The EMC filter of claim 5, wherein a thickness of the first bus bar is different from a thickness of the second or third bus bar in the first direction.
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